# Progressive block-size sweeps on QuickDraw

Five families × two inits × four rates ($m = n = 5$, $32 \times 32$)

Mean test PSNR (dB) over 50 drawings. n/a = stage undefined (mera needs $k$ a power of 2).

## 1 Low-resolution drawings

QuickDraw is $32 \times 32$ ($m = n = 5$; $k = 1..5$, shown $k = 2..5$, `mera` at $k \in \{2, 4\}$). At this low resolution $8 \times 8$-block DCT is weak ($\approx 26.6$ dB @ $\rho = 0.20$), so the learned circuits beat it.

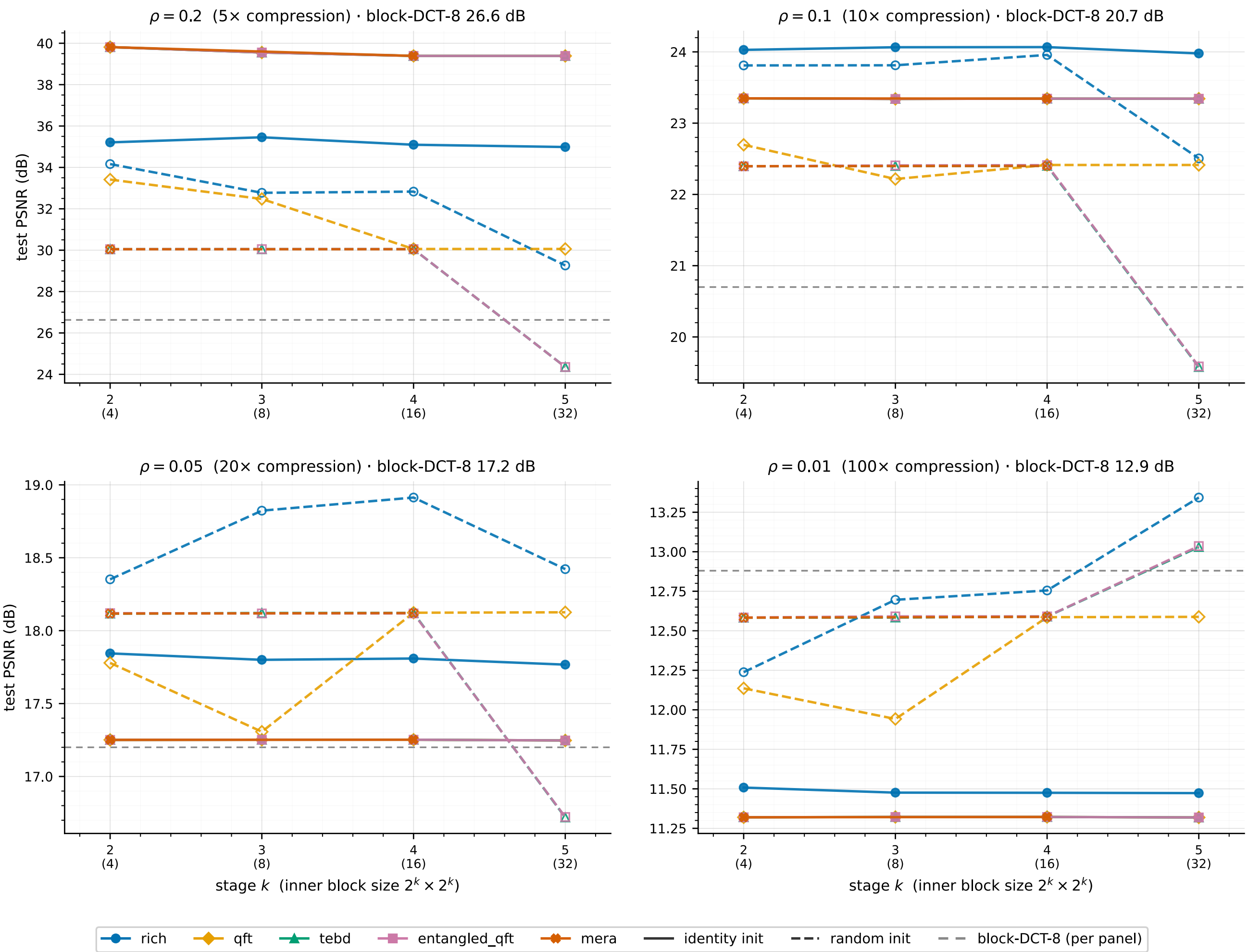


Figure 1: PSNR vs block size $k$ at four keep ratios. One colour per family, solid = identity / dashed = random init; grey dashed = block-DCT-8 (value per panel).

## 2 Observations

- **Learned circuits beat block-DCT at every rate** (unlike the $256^2$ sets).
- **The QFT-derived families lead and are bit-identical under identity init,** $\approx 4$ dB above `rich` @ $\rho = 0.20$; curve flat in $k$ (like DIV2K).
- **At heavy compression families converge** to block-DCT and the identity advantage erodes (random sometimes edges ahead).

## 3 Per-rate tables (mean PSNR, dB)

$\rho = 0.20$ — **5×**

| family | init | $k$=2\ (4) | $k$=3\ (8) | $k$=4\ (16) | $k$=5\ (32) |
|---|---|---|---|---|---|
| rich | identity | 35.2 | 35.5 | 35.1 | 35 |
| rich | random | 34.2 | 32.8 | 32.8 | 29.3 |

| family | init | $k$=2\ (4) | $k$=3\ (8) | $k$=4\ (16) | $k$=5\ (32) |
|---|---|---|---|---|---|
| qft | identity | 39.8 | 39.6 | 39.4 | 39.4 |
| qft | random | 33.4 | 32.5 | 30.1 | 30.1 |
| tebd | identity | 39.8 | 39.6 | 39.4 | 39.4 |
| tebd | random | 30 | 30 | 30 | 24.4 |
| ent-qft | identity | 39.8 | 39.6 | 39.4 | 39.4 |
| ent-qft | random | 30 | 30.1 | 30.1 | 24.4 |
| mera | identity | 39.8 | n/a | 39.4 | n/a |
| mera | random | 30 | n/a | 30 | n/a |

$\rho = 0.10$ — **10**×

| family | init | $k$=2\ (4) | $k$=3\ (8) | $k$=4\ (16) | $k$=5\ (32) |
|---|---|---|---|---|---|
| rich | identity | 24 | 24.1 | 24.1 | 24 |
| rich | random | 23.8 | 23.8 | 24 | 22.5 |
| qft | identity | 23.3 | 23.3 | 23.3 | 23.3 |
| qft | random | 22.7 | 22.2 | 22.4 | 22.4 |
| tebd | identity | 23.3 | 23.3 | 23.3 | 23.3 |
| tebd | random | 22.4 | 22.4 | 22.4 | 19.6 |
| ent-qft | identity | 23.3 | 23.3 | 23.3 | 23.3 |
| ent-qft | random | 22.4 | 22.4 | 22.4 | 19.6 |
| mera | identity | 23.3 | n/a | 23.3 | n/a |
| mera | random | 22.4 | n/a | 22.4 | n/a |

$\rho = 0.05$ — **20**×

| family | init | $k$=2\ (4) | $k$=3\ (8) | $k$=4\ (16) | $k$=5\ (32) |
|---|---|---|---|---|---|
| rich | identity | 17.8 | 17.8 | 17.8 | 17.8 |
| rich | random | 18.4 | 18.8 | 18.9 | 18.4 |
| qft | identity | 17.3 | 17.3 | 17.3 | 17.2 |
| qft | random | 17.8 | 17.3 | 18.1 | 18.1 |
| tebd | identity | 17.3 | 17.3 | 17.3 | 17.2 |
| tebd | random | 18.1 | 18.1 | 18.1 | 16.7 |
| ent-qft | identity | 17.3 | 17.3 | 17.3 | 17.2 |
| ent-qft | random | 18.1 | 18.1 | 18.1 | 16.7 |
| mera | identity | 17.3 | n/a | 17.3 | n/a |
| mera | random | 18.1 | n/a | 18.1 | n/a |

$\rho = 0.01$ — **100**×

| family | init | $k$=2\ (4) | $k$=3\ (8) | $k$=4\ (16) | $k$=5\ (32) |
|---|---|---|---|---|---|
| rich | identity | 11.5 | 11.5 | 11.5 | 11.5 |
| rich | random | 12.2 | 12.7 | 12.8 | 13.3 |
| qft | identity | 11.3 | 11.3 | 11.3 | 11.3 |
| qft | random | 12.1 | 11.9 | 12.6 | 12.6 |
| tebd | identity | 11.3 | 11.3 | 11.3 | 11.3 |
| tebd | random | 12.6 | 12.6 | 12.6 | 13 |
| ent-qft | identity | 11.3 | 11.3 | 11.3 | 11.3 |
| ent-qft | random | 12.6 | 12.6 | 12.6 | 13 |
| mera | identity | 11.3 | n/a | 11.3 | n/a |
| mera | random | 12.6 | n/a | 12.6 | n/a |